

5 (long) A non-inverting Operational amplifier circuit is shown in Fig. 5a.

(a) Derive an expression for the voltage gain of the circuit, stating any assumptions made. [6]

(b) Two capacitors, C_1 and C_2 are now added as shown in Fig. 5b, where $C_1 \gg C_2$. Derive an expression for the voltage gain of this circuit as a function of frequency and sketch what its magnitude looks like, clearly labelling the salient features. [12]

(c) The circuit in Fig. 5b is required to have a mid-band voltage gain of 100 and upper and lower -3 dB frequencies of approximately 1 MHz and 100 Hz, respectively. Determine appropriate values for R_2 , C_1 and C_2 assuming that $R_1 = 5 \text{ k}\Omega$. State any approximations made. [10]

(d) For the case where R_2 is replaced with an inductor, briefly describe how you expect the gain to vary as a function of frequency. [2]

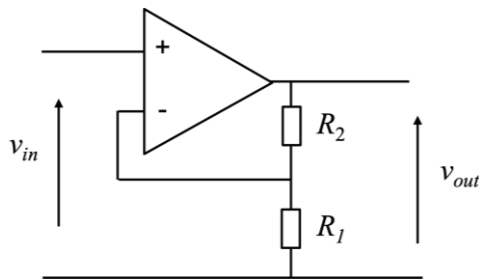


Fig. 5a

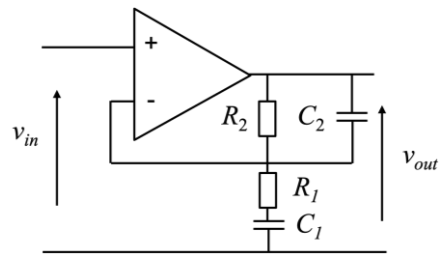


Fig. 5b